

Nathan S. Martin

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EDUCATION

Bachelor of Software Engineering | University of Waterloo

Sep 2023 – Apr 2028

- Joint Honours in Combinatorics and Optimization | Artificial Intelligence Specialization | Physical Sciences Option
- **97.2%** Major Average | **96.0%** Cumulative Average | **4.0** GPA

EXPERIENCE

Roblox | Systems Software Engineer Intern

San Mateo, California | May 2026 - Aug 2026

- Drove deployment and integration of a custom, high-performance cooperative yielding runtime environment
- Engineered a fiber runtime debugging tool supporting synchronization arcs, wait-for graphs, and deadlock detection
- Developed highly optimized systems code, reducing memory overhead and maximizing runtime execution speed

BitGo | Software Engineering Intern

Palo Alto, California | Sep 2025 - Dec 2025

- Led backend for a BTC yield product via Lightning Network node orchestration, liquidity leasing, and routing fees
- Engineered secure blockchain infrastructure by automating node setups, Macaroon auth, and transaction tracking
- Designed optimized SQL schemas and TypeScript APIs to process, store, and serve transaction and yield data to the UI

University of Waterloo | Undergraduate Research Assistant

Waterloo, Ontario | Jan 2025 – Present

- **Quantum Algorithms** | Dr. Ashwin Nayak: Proving optimal bounds for classes of algorithms for quantum computers
- **Formal Verification** | Dr. Nancy Day: Developed a formal verification tool for validating first-order logic and program correctness. Co-authoring a research paper on an LLM benchmark for the formal modelling language Alloy

Ford Motor Company | Automotive Software Developer

Waterloo, Ontario | Jan 2025 - Apr 2025

- Designed asynchronous C++ hardware abstraction layers (HALs) for real-time control logic and ECU communication
- Developed production-grade concurrent software, ensuring reliability and performance across millions of active vehicles

Connect Tech Inc | Embedded Software Developer

Guelph, Ontario | May 2024 - Aug 2024

- Developed BSPs for Nvidia Jetson carrier boards by writing kernel drivers, device trees, and shell scripting automation
- Implemented disk encryption and OTA updates, enhancing security and serviceability of devices deployed in the field

McMaster University | Computational Physics Developer

Hamilton, Ontario | Feb 2023 - Jun 2023

- Developed force and impulse-based, real time, rigid body physics simulations using Panda3D and Python
- Designed and implemented algorithms to triangulate, render, and compute collisions of prisms and ellipsoids

PROJECTS

Skatelligence | AI-Powered Figure Skating Analysis

[Website](#)

- Created an LSTM-based recurrent neural network (RNN) with PyTorch to classify figure skating jumps with 80% accuracy
- Built a generative adversarial network (GAN) to generate synthetic IMU data, improving identification accuracy by 50%
- Programmed an ESP32 microcontroller to use a RTOS for multithreaded IMU data collection and transmission to a server

Waterloo Aerial Robotics Group

- Leveraged PyTorch and YOLOv8 classification model to extract landing pad locations from real-time video
- Engineered the autopilot decision-making algorithm, integrating telemetry, computer vision data, and mission commands

Rubik's Cube Solving Robot

[Website](#)

- Developed an optimized C++/CUDA algorithm for a custom robot to solve cubes in half the moves of top speedcubers

SKILLS

Languages: C++, C, Python, Java, Scala, Kotlin, Shell Script, C#, VHDL, SQL, JavaScript / TypeScript

Technologies: CUDA, PyTorch, OpenCV, AForge.NET, Git, SVN, MongoDB, Firebase, Docker, REST APIs, LaTeX, Fusion360, Raspberry Pi, Arduino, ESP32, Nvidia Jetson, KiCad, FreeRTOS